

Peijin Wang

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Research Interests

Bayesian methods for clinical trials; causal inference; external controls and dynamic information borrowing; adaptive clinical trial design; real-world data and real-world evidence

Education

Duke University

PHD IN BIOSTATISTICS

Durham, NC

Expected May 2027

- Thesis: Innovative Bayesian Statistical Approaches in Clinical Trials Leveraging External Controls
- Advisor: Hwanhee Hong, PhD
- Committee: Laine E. Thomas, PhD; Sean M. O'Brien, PhD; Samuel I. Berchuck, PhD; Scott M. Palmer, MD

Duke University

MASTER OF BIOSTATISTICS

Durham, NC

May 2022

- Thesis: The Impact of Block Size on Treatment Imbalance and Blinding Integrity in Randomized Clinical Trials
- Advisor: Shein-Chung Chow, PhD

Central South University

BS IN STATISTICS

Changsha, China

June 2020

- Thesis: Connectedness and Hedging between Hedging Assets and Stock Markets
- Advisor: Hongwei Zhang, PhD

Professional Experience

Research Assistant, Duke University

Aug. 2024 - Present

PI: Hwanhee Hong, PhD

Durham, NC

- Collaborate with clinicians on statistical analyses addressing clinical questions in atrial fibrillation using the COMBINE AF database.
- Develop and evaluate methods for leveraging external control data from chronic lung allograft dysfunction (CLAD) cohorts to inform future hybrid controlled trial designs.

Statistical Scientist Intern, Berry Consultants

May 2025 - Aug. 2025

Main Project: Leveraging External Controls in Amyotrophic Lateral Sclerosis Single-Arm Trials

Remote

- Lead simulation studies evaluating causal inference methods for leveraging external controls in small single-arm amyotrophic lateral sclerosis (ALS) trials.
- Contribute to Bayesian clinical trial design and simulation involving external data borrowing.

Research Assistant, Duke University

Sep. 2021 - May 2022

PI: Shein-Chung Chow, PhD

Durham, NC

- Collaborate with clinical investigators on statistical analyses for an observational study of rehabilitation in lung transplantation.
- Support the design and analysis of a randomized palliative care trial.

Publications

PEER-REVIEWED JOURNAL ARTICLES

1. Wang, P., Hong, H., Jeon, K., & Thomas, L. (2026). Trials augmented by external control data using balancing weights: a comparison of estimands and estimators. *Contemporary Clinical Trials*, 108220.
2. Zhang, Y., Zhang, X., Wang, P., Wu, Y., & Chow, S. C. (2025). A proposed confidence ellipse approach for benefit-risk assessment in clinical trials. *Therapeutic Innovation & Regulatory Science*, 1–13.
3. Wang, P., & Chow, S. C. (2024). The use of real-world data for clinical investigation of effectiveness in drug development. *Journal of Biopharmaceutical Statistics*, 1–24.
4. Chow, S. C., & Wang, P. (2024). On the use of RWD in support of regulatory submission in drug development. *Journal of Biopharmaceutical Statistics*, 1–28.
5. Byrd, R., Breslin, R., Wang, P., Peskoe, S., Chow, S. C., Lowers, S., Snyder, L. D., & Pastva, A. M. (2024). Group versus individual rehabilitation in lung transplantation: A retrospective noninferiority assessment. *Cardiopulmonary Physical Therapy Journal*, 35(1), 28–36.

6. Gu, J., **Wang, P.**, Chow, S. C., Dempsey, K., Bermejo, S., Swaminathan, A., Soskis, A., Fried, J., Kloefkorn, C., & Jones, C. (2024). An App platform-facilitated collaborative palliative care intervention for outpatients with interstitial lung disease: A pilot randomized trial. *American Journal of Hospice and Palliative Medicine*®, 10499091241275966.
7. **Wang, P.**, & Chow, S. C. (2023). Innovative thinking of clinical investigation for rare disease drug development. *Orphanet Journal of Rare Diseases*, 18(1), 299.
8. **Wang, P.**, & Chow, S. C. (2022). The impact of block size on treatment imbalance and blinding integrity in randomized clinical trials. *Authorea Preprints*.
9. **Wang, P.**, Peskoe, S., Byrd, R., Smith, P., Breslin, R., & Chow, S. C. (2022). Statistical evaluation of absolute change versus responder analysis in clinical trials. *Acta Materia Medica*, 1(3), 320.
10. **Wang, P.**, & Chow, S. C. (2021). Sample size re-estimation in clinical trials. *Statistics in Medicine*, 40(27), 6133–6149.
11. **Wang, P.**, Zhang, H., Yang, C., & Guo, Y. (2021). Time and frequency dynamics of connectedness and hedging performance in global stock markets: Bitcoin versus conventional hedges. *Research in International Business and Finance*, 58, 101479.
12. Zhang, H., & **Wang, P.** (2021). Does Bitcoin or gold react to financial stress alike? Evidence from the US and China. *International Review of Economics & Finance*, 71, 629–648.

BOOK CHAPTERS

13. Song, F., Bai, H., **Wang, P.**, & Chow, S. C. (2021). Non-Medical switch. In *Methodologies in Biosimilar Product Development* (pp. 277–290). Chapman and Hall/CRC.
14. Zheng, J., **Wang, P.**, Wang, Y., & Chow, S. C. (2021). Detecting reference product change in biosimilar studies. In *Methodologies in Biosimilar Product Development* (pp. 227–251). Chapman and Hall/CRC.

UNDER REVIEW AND IN PREP

15. **Wang, P.**, ... & Hong, H. Integrating multiple external controls with randomized controlled trial data in treatment effect evaluation: a case study using CLAD cohort data.
16. **Wang, P.** & Hong, H. Innovative Bayesian dynamic borrowing methods for leveraging multiple external controls in randomized controlled trials.
17. Yanchenko, E., **Wang, P.**, ... & Hong, H. Effect of imbalanced treatment allocation ratio on combining multiple historical controls in clinical trials.
18. Vestrucci, M.*, **Wang, P.***, ... & Quintana, M. External controls for exploratory efficacy assessment in early-phase ALS trials: A PRO-ACT–Based simulation study and illustrative case study. (*: equal contribution)
19. Xie, A., ..., **Wang, P.** & Ma, C. A novel sequential multiple assignment randomized trial design with internal pilot study and unblinded sample size re-estimation (SMARTIES).
20. Neely, M., Wojdyla, D., Hong, H., **Wang, P.**, ..., Todd, J. Latent class analysis identifies pulmonary function trajectory phenotypes in lung transplant recipients with chronic allograft dysfunction.
21. Reddy, R., Lydon, E., Lopes, R., **Wang, P.**, ... & Granger, C. Gastrointestinal bleeding in atrial fibrillation: a patient-Level pooled analysis of large-scale randomized trials from COMBINE AF.
22. Hughes, S., Tannu, M., **Wang, P.**, ... & Pokorney, S. INR values in patients on Apixaban: results from the ARISTOTLE trial.
23. Tannu, M., Hua, K., **Wang, P.**, ... & Granger, C. Predictors and associated outcomes of intracranial hemorrhage in patients with atrial fibrillation on oral anticoagulation: Insights from COMBINE-AF.

Honors & Awards

TRAVEL AND COMPETITION AWARDS

- 2026 **ASA BIOP Scholarship Award**, American Statistical Association Biopharmaceutical Section
- 2025 **Student Travel Award**, American Statistical Association Biopharmaceutical Section Regulatory-Industry Statistics Workshop (RISW)
- 2025 **Best Poster Award**, Duke Industry Statistical Symposium (DISS)
- 2025 **Student Travel Award**, North Carolina Chapter of the American Statistical Association
- 2025 **Student Paper Travel Award**, International Conference on Health Policy Statistics (ICHPS)
- 2019 **First Prize**, China University Student Statistical Modeling Competition
- 2018 **Honorable Mention Award**, Mathematical Contest in Modeling/Interdisciplinary Contest in Modeling Contest

DUKE UNIVERSITY

- 2023 International Chancellor's Scholarship
- 2022 Overall Academic Achievement of Duke Master of Biostatistics Program
- 2022 Student Leadership Awards Duke Master of Biostatistics Program
- 2022 Honorable Mentions for the Masters Project

CENTRAL SOUTH UNIVERSITY

2017-19 3rd Scholarship, Academic Excellence Award

Presentations

ORAL PRESENTATIONS

- 2026 A Bayesian Adaptive Design with Sample Size Re-estimation for Hybrid Controlled Trials.
Joint Statistical Meetings (JSM). August, Boston, MA.
- 2026 A Bayesian Adaptive Design with Sample Size Re-estimation for Hybrid Controlled Trials.
Duke Industry Statistical Symposium (DISS). April, Durham, NC.
- 2025 Innovative Bayesian Dynamic Borrowing Methods for Leveraging Multiple External Controls in Randomized Controlled Trials.
New England Statistics Symposium. June, New Haven, CT.
- 2025 Innovative Bayesian Dynamic Borrowing Methods for Leveraging Multiple External Controls in Randomized Controlled Trials.
Eastern North American Regional (ENAR) Meeting of the International Biometric Society. March, New Orleans, LA.
- 2022 Statistical evaluation of absolute change versus responder analysis in clinical trials.
Annual Duke Transplant Center IT Symposium. September, Virtual.

POSTER PRESENTATIONS

- 2025 Innovative Bayesian Dynamic Borrowing Methods for Leveraging Multiple External Controls in Randomized Controlled Trials.
ASA Biopharmaceutical Section Regulatory-Industry Statistics Workshop (RISW). September, Rockville, MD.
- 2025 Innovative Bayesian Dynamic Borrowing Methods for Leveraging Multiple External Controls in Randomized Controlled Trials. (Poster Lightning Talk)
Duke Industry Statistical Symposium (DISS). April, Durham, NC.
- 2025 Integrating Randomized Controlled Trial and External Control Data Using Balancing Weights: A Comparison of Estimands and Estimators.
International Conference on Health Policy Statistics (ICHPS). January, San Diego, CA.
- 2024 The Use of Real-World Data for Clinical Investigation of Effectiveness in Drug Development.
Eastern North American Regional (ENAR) Meeting of the International Biometric Society. March, Baltimore, MD.

Teaching

- 2022-23 Generalized Linear Models (BIOS 719), Department of Biostatistics and Bioinformatics, Duke University.
Instructors: Hwanhee Hong, PhD.
Responsibilities: grading and holding office hours.
- 2023-24 Clinical Trial Design and Analysis (BIOS 718), Department of Biostatistics and Bioinformatics, Duke University.
Instructors: Xiaofei Wang, PhD.
Responsibilities: writing homework solutions, grading, and holding office hours.

Professional Activities

SESSION CHAIR

- 2026 Joint Statistical Meetings. *August, Boston, MA.*

VOLUNTEER

- 2025-26 Student Council Member. *Duke Biostatistics Student Council.*

- 2025 Student Coordinator. *Duke University BIOS 900, a seminar course for more than 30 Biostatistics PhD students.*
- 2025 Student Volunteer. *Duke Industry Statistics Symposium. April, Durham, NC.*
- 2024 Student Volunteer Lead. *Duke Industry Statistics Symposium. April, Durham, NC.*

SOCIETY MEMBER

- 2022- American Statistical Association (ASA)
- 2022- Eastern North American Region International Biometric Society (ENAR)
- 2022- International Biometric Society (IBS)

REVIEWERS

Journals *Acta Materia Medica* (1), *Orphanet Journal of Rare Diseases* (1), *PLOS ONE* (1), *Drug Design, Development and Therapy* (1), *Journal of Pharmaceutical Policy and Practice* (2), *Clinical Trials* (1), *Pharmacology Research & Perspectives* (1), *Statistics in Biopharmaceutical Research* (2), *Pharmaceutical Statistics* (1), *Journal of Biopharmaceutical Statistics* (1)

Computing Skills

- Programming R, JAGS, Stan, SAS
- Operating Systems Unix/Linux, Windows